Naman Chetwani

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PROJECTS

wrdn

- Building **wrdn**, an AI accountability system that watches the day through ambient audio and visual context, models goals and values over time, and flags drift from what actually matters.
- Designed the memory layer, personal-agent command execution, and flexible hardware strategy across a clip-on device, Meta Ray-Bans, and Brilliant Labs Halo.
- Building the stack **from kernels to hardware**, with an auditable core serious users can inspect instead of trusting branding.

Gemma 4 Inference Megakernel

- Building **gemma.c**, an experimental Gemma 4 dense inference runtime: standalone CUDA/PTX kernels first, then a correct unfused baseline and measured fusion toward a specialized megakernel path.

STRATUS UAV: Optimizing Prescribed Burning using UAVs and Predictive Modeling

- Led development of a **patent-pending autonomous UAV** to mitigate wildfire spread; 3D-printed a VTOL aircraft for autonomous backburning with **15 km range** and **1.2 hour endurance**.
- Targeted core field constraints: helicopter operations around $16k/day, custom aircraft and pilots, and pilot liability; designed for **5x lower price** and **3x higher range** than the closest drone competitor.

SBON (Strategic-Blackline-Optimization-Engine)

- Built a wildfire prediction foundational model that **broke SOTA** and created the first ML approach to optimizing prescribed burns.
- Built a tri-branch multimodal architecture across **13 input modalities** spanning fire imagery, weather, and geography; ConvNeXt and SSM encoders feed a larger SSM for 4-hour fire-mask prediction.
- Treated the model as an environment, rolled out predictions to **48 hours**, and used PPO to optimize burn placement in a niche field with fewer than 20 papers in the last 30 years.

No Circles

no-circles.com

- Built a website-first personalized daily newsletter that learns from what each reader responds to; scaled to **triple-digit users 3 weeks from inception**.
- Built the memory, brief, and feedback loop from replies and user signals to teach people about tangential fields they know little about.

PHOENIX UAV / Science Fair Results

- Built PHOENIX UAV for early wildfire detection; placed **2nd in the California State Science Fair**, **1st at regionals**, and won 5 awards including the Henkel aerospace award and Junior Lemelson-MIT Inventor Award.

WORK

Fire Technology Contract Work

Hardware / Systems Engineer

- Under contract building machines for fire tech companies, combining autonomous hardware, manufacturing, and field-oriented fire systems.

EDUCATION

Independent Study / Competitions

- Independently studied mathematics from high school precalculus to multivariable calculus and linear algebra, plus ML and transformer theory, in under a year.
- Selected for **SPARC '26** (3% acceptance rate); achieved **USACO Gold** and F=MA qualifying scores in the top 5% nationally.
- Competed in CALICO, Harker Physics Invitational, and Harker Programming Invitational; held a top 30 national placement in MSPF Debate with elimination rounds at Stanford and TOC.